CARLOS ARIEL ORTEGA GUERRON

Data Scientist | Biomedical Engineer

Quito, Ecuador | +593 995517709 | arielguerron@gmail.com | LinkedIn: https://www.linkedin.com/in/ariel-ortega-guerron/ | GitHub: https://github.com/ArielOGc1

PROFESSIONAL PROFILE

Data Scientist with training in Biomedical Engineering specializing in the development of analytical solutions under the Python and SQL ecosystem. I have technical expertise in Linux environments, version control systems, and the deployment of efficient algorithms for solving complex problems. I have experience in the technical-biomedical sector, with strong statistical analysis skills and the ability to execute precision technical protocols. I stand out for my self-taught approach focused on process optimization and continuous learning in technological environments.

TECHNICAL KNOWLEDGE

Python | SQL | Plotly | Streamlit | Seaborn | NumPy | Matplotlib | Scikit-learn | Pandas | GitHub | TensorFlow | Keras | Git | Bash | Linux | Excel | Power BI

PROFESSIONAL EXPERIENCE

Technical Assistant (Freelance), Dental Medical Solutions | September 2024 – October 2025

• Extended the operational life of critical equipment by more than 20% by identifying and replacing defective components in suction systems.

• Ensured 100% compliance with technical and safety protocols in all interventions performed.

• Resolved more than 90% of technical incidents on the first intervention, minimizing interruptions in clinical care.

• Optimized the preventive maintenance process, reducing recurring failures in serviced equipment by approximately 25%.

Technical Volunteer, Biomedicine Masters | February 2024 – September 2025

• I supported the technical organization of training programs in biomedical equipment, contributing to the training of more than 30 professionals in the health sector.

• Designed and implemented a basic inventory control system for more than 50 materials and devices used in the courses, improving the organization and availability of technical resources.

• Ensured 100% operability of medical devices before each practical session through technical reviews and initial calibrations.

PROJECTS

Credit Risk Classification System (Credit Scoring) | 2026

- Description: Development of a predictive model to assess the probability of default and creation of a web interface for viewing financial profiles.
- Technologies Used: Python, Pandas, Scikit-learn, Streamlit, Matplotlib, NumPy.
- Key Responsibilities: Financial data cleaning, handling outliers, and applying resampling techniques for class imbalance.
- Impact/Achievements: Translation of the analytical model into a functional tool facilitating the interpretation of metrics (F1-Score, ROC-AUC).

Selection of Regions for Oil Extraction | 2025

- Description: Regression model to optimize the location of new wells based on reserve volume and ROI.
- Technologies Used: Python, Scikit-learn, Pandas, Bootstrapping.
- Key Responsibilities: Development of linear regression and execution of Bootstrapping (1,000 iterations) for risk assessment.
- Impact/Achievements: Identification of region with risk of loss less than 2.5% and estimated profitability of $3.35 million USD.

Prediction of Phone Plans | 2025

- Description: Analysis of user behavior for recommending phone plans through classification.
- Technologies used: Python, Scikit-learn, Pandas, Matplotlib.
- Key responsibilities: Database consolidation and model training (decision trees, random forest).
- Impact/achievements: 75% accuracy, enabling commercial strategies to be targeted.

EDUCATION

Universidad Politécnica Salesiana | Ingeniería Biomédica (In progress)

TripleTen (Bootcamp Online) | Data Scientist (2025)

Udemy | Especialización en Programación con Python (2025)

Udemy | Algoritmos y Estructuras de Datos (In progress)

LANGUAGES

Spanish (Native) | English (B2)